PROTOTYPE DEMONSTRATION GUIDE

Enhancing Kenya Power's Prepaid Electricity Experience Through Bluetooth-Enabled Smart Customer Interface Unit (CIU) Integration

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1. Purpose of the Prototype

The purpose of this prototype is to demonstrate the feasibility of securely integrating a Bluetooth-enabled communication module with an existing Customer Interface Unit (CIU) to improve customer interaction with Kenya Power's prepaid electricity ecosystem.

The prototype is intended to validate the concept and demonstrate how customers can interact with their CIU through the official MyPower mobile application while preserving the current prepaid infrastructure.

This prototype is **not intended to replace existing CIUs or prepaid meters**, but rather to illustrate how a communication layer could enhance the current customer experience.

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2. Prototype Objectives

The prototype demonstrates the following capabilities:

- Secure Bluetooth pairing between a mobile device and the communication module.
- Two-way communication between the mobile application and the communication module.
- Automatic transmission of prepaid electricity tokens from the mobile application to the Customer Interface Unit.
- Retrieval of electricity information available through the Customer Interface Unit.
- Mobile display of electricity information.
- Backward compatibility with existing manual token entry.

3. Prototype Components

The prototype consists of the following major components:

Mobile Application

Simulates the customer experience through an Android application capable of securely communicating with the Bluetooth communication module.

Functions include:

- Device pairing
- Token transmission
- Information display
- Notification handling

Bluetooth Communication Module

Acts as an intermediary between the mobile application and the Customer Interface Unit.

Responsibilities include:

- Secure Bluetooth communication
- Command processing
- Data exchange
- Communication management

Customer Interface Unit (CIU)

Represents the existing prepaid Customer Interface Unit.

The communication module interfaces with the CIU to:

- Receive customer commands
- Retrieve electricity information
- Transfer prepaid tokens

4. Demonstration Workflow

The demonstration follows the sequence below:

Step 1

Launch the mobile application.

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Step 2

Pair the application with the Bluetooth communication module.

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Step 3

Establish a secure Bluetooth connection.

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Step 4

Request electricity information from the Customer Interface Unit.

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Step 5

Display electricity information on the mobile application.

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Step 6

Simulate the purchase of a prepaid electricity token.

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Step 7

Transmit the token securely to the communication module.

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Step 8

The communication module forwards the token to the Customer Interface Unit.

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Step 9

The Customer Interface Unit processes the token.

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Step 10

The mobile application confirms successful transmission.

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5. Current Prototype Capabilities

The prototype successfully demonstrates:

✔ Secure Bluetooth communication

✔ Mobile application pairing

✔ Two-way data exchange

✔ Automatic token transmission

✔ Mobile monitoring

✔ Real-time communication

✔ Concept validation

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6. Current Limitations

As a proof of concept, the prototype has the following limitations:

- Developed within a controlled demonstration environment.
- Uses simulated integration where production interfaces are unavailable.
- Does not modify Kenya Power's existing prepaid infrastructure.
- Full production integration would require collaboration with Kenya Power and Customer Interface Unit manufacturers.

7. Future Development

Future collaboration may include:

- Integration with production Customer Interface Units.
- Hardware optimization.
- Security certification.
- Field trials.
- Pilot deployments.
- Customer usability evaluation.

Conclusion

The prototype demonstrates that secure Bluetooth communication can enhance customer interaction with existing prepaid electricity infrastructure while maintaining compatibility with current operational processes.

The proof of concept provides a foundation for further technical evaluation and collaborative development.